

Time-of-Flight Mass Spectrometer

Demo 13 of 16 · Part four, Example projects · 12–15 minutes for stages 1–3, plus about 5 for stage 4

You need: `index.html` open in any modern browser, on a projector or on each student's phone. No network, no install, no accounts. Nothing is fetched at runtime, so it works with the wi-fi off.

Learning goals

One per stage, in the learner's own words — these are the four Takeaway cards on screen, verbatim. If a student can say these four things and point at the evidence for each, the demo has done its job.

1. Flight time grows with the square root of mass, not with mass: four times the mass takes twice the time.
2. The same voltage gives a doubly charged ion twice the energy, so it crosses the tube sooner even at identical mass.
3. Flight time fixes m/z and nothing more, so one peak does not establish one kind of ion.
4. Whether a longer tube separates two peaks depends entirely on how the peak width behaves — the length alone does not decide it.

Where this sits on the path

An AI-built classroom tool from Part four. Its goal is to separate what a detector measures from what you may conclude, its engine is live on the device, and its assumptions are one tap away in **Details**. It does not depend on any earlier demo, so it can be taught on its own; the habit it shares with the rest of Part four is naming the boundary between mechanism and claim.

One question for the session

What does a flight time actually tell you about an ion?

"This instrument measures one thing: how long an ion took to cross a tube. Everything else we are about to say is an inference from that one number. Our job for the next fifteen minutes is to find out exactly how far that number gets us — and where it stops."

Units students see

Label	Meaning in this activity
dalton (Da)	A mass unit used for atoms and molecules.
elementary charge (e)	+1e or +2e is positive charge. The dimensionless charge number z is 1 or 2.
microsecond (μs)	One millionth of a second. The detector clock and the trace's horizontal axis show real flight time in μs .
metre (m)	Flight-tube length. Stages 1–3 hold it at 1.50 m; stage 4 compares 1.00 m with 2.00 m.
kilovolt (kV)	One thousand volts. Every stage holds the boost at 20 kV.

The screen uses one scale for every core run: 1.00 physical μs takes 160 ms on screen, or 160,000× slow motion. The detector trace keeps the same 0–18 μs horizontal axis in all three core stages, so runs can be compared by eye.

Stage 1 · Change mass

Question: Both ions carry the same single positive charge. Which one reaches the detector first, and by how much?

Predict prompt: Which packet arrives first — 100 Da or 400 Da, both at +1? Run is locked until a student commits, and the list locks once the detector has answered. Take a show of hands before anyone taps.

Activity (3 minutes): 0:00 read the two ion cards and the two fixed conditions · 0:30 commit a prediction · 0:45 press **Run detector** · 1:00 watch both markers cross, and read the cue line under the flight window as each arrives · 2:00 read the finding and the summed trace.

Expected observation: Ion A at **7.64 μ s**, Ion B at **15.27 μ s**; separation 7.64 μ s. Exactly a factor of two, which is the square root of the four-fold mass difference.

Check-yourself answer key: "Twice as long." The tempting wrong answer is "four times as long, because it is four times the mass" — that is the linear-in-mass error, and the observed 15.27 μ s refutes it directly. The third option denies mass matters at all; the two traces refute it.

Teaching limit: Do not let "lighter is faster" become "flight time measures mass." Stage 2 breaks that immediately, and stage 1 is where students build the habit of saying what was held constant.

Misconception: flight time is proportional to mass. It is proportional to the square root of mass, because the voltage fixes the ion's energy, not its speed. Equal energy means a four-times-heavier ion moves half as fast, not a quarter as fast. Point at 7.64 and 15.27.

Stage 2 · Change charge

Question: Both ions weigh 200 Da. If one carries twice the charge, what does the same voltage do to it?

Predict prompt: Two ions of 200 Da, one at +1 and one at +2 — which arrives first? This one splits a room, and the split is worth surfacing before the run.

Activity (3 minutes): 0:00 note that both cards now read 200 Da and only the charge differs · 0:30 commit · 0:45 **Run detector** · 1:30 read the two arrivals against the prediction echo in the Predict card.

Expected observation: Ion A (+1) at **10.80 μ s**, Ion B (+2) at **7.64 μ s**. The ratio is 1.414 — the square root of two — because the +2 ion gained twice the kinetic energy.

Check-yourself answer key: "The same voltage gave it twice the kinetic energy." The two wrong options are the ones students actually reach for: that a +2 ion is somehow lighter (both cards say 200 Da), and that the tube keeps pulling on it during flight (the tube is field-free — the whole difference was made before the clock started).

Teaching limit: The demo shows that charge changes flight time in an ideal instrument. It does not show that charge states are easy to assign in a real spectrum.

Misconception: the drift tube accelerates the ions. It does not — that is what "field-free" means, and it is why the clock can start at the tube entrance. All the work was done by the voltage before timing began. If the tube were still pushing, doubling its length would not simply double the flight time.

Stage 3 · Match the ratio

Question: Ion A is 200 Da at +1 and Ion B is 400 Da at +2. What does the detector record?

Predict prompt: Equal m/z , different mass and charge — one peak or two? Ask for the reasoning before the vote; this is the stage where the reasoning matters more than the answer.

Activity (3 minutes): 0:00 compute $200 \div 1$ and $400 \div 2$ together on the board · 0:45 commit · 1:00 **Run detector** · 2:00 look at where the two coloured centre ticks land on the trace, and at the single peak the summed signal makes.

Expected observation: Both centres at **10.80 μ s**. The page checks the two times for exact equality, so the readout says "coincident arrival" rather than "close".

Check-yourself answer key: "Different ions could share the same mass-to-charge ratio." The wrong option — "exactly one kind of ion must be present" — is the conclusion a real spectrum invites and this stage has just refuted. Sit on this one; it is the demo's target claim.

Teaching limit: The claim is narrow and it is the important one: a single flight time constrains m/z and nothing else. It is *not* the claim that real identification is impossible — only that this measurement alone cannot do it.

Misconception: one peak means one substance. One peak means one m/z . Two genuinely different ions landed on the same tick in front of the class. Real identification brings isotope patterns, fragment spectra, chromatographic separation and accurate-mass calibration — none of which exist on this page, which is exactly why flight time alone is not enough.

Stage 4 · Peak width

Question: A 1.00 m tube cannot tell 200 Da from 205 Da. Does making the tube longer fix that?

Predict prompt: Doubling the tube doubles the gap between two arrivals — is that enough to separate the peaks? Most students say yes, and under one of the two assumptions they are right, which is what makes the stage work.

Activity (5 minutes): 0:00 introduce *timing width* in words before naming sigma · 1:00 **Run paired comparison** on the fixed-width assumption and read the table · 2:30 switch to **Proportional width** and run it again · 4:00 compare the two separation-to-width ratios.

Assumed width rule	1.00 m result	2.00 m result	What changes
Fixed $\sigma = 0.06 \mu\text{s}$	Separation $0.089 \mu\text{s}$ FWHM $0.141 \mu\text{s}$ ratio 0.633 · overlapping	Separation $0.179 \mu\text{s}$ FWHM $0.141 \mu\text{s}$ ratio 1.266 · resolved	Separation doubles while the absolute width stays fixed.
$\sigma = 1.5\%$ of arrival time	Separation $0.089 \mu\text{s}$ mean FWHM $0.256 \mu\text{s}$ ratio 0.350 · overlapping	Separation $0.179 \mu\text{s}$ mean FWHM $0.512 \mu\text{s}$ ratio 0.350 · overlapping	Separation and width both double; the ratio does not move.

Check-yourself answer key: "No — it depends on how the peak width behaves." The wrong options are "a longer path always separates peaks better" (the proportional run refutes it on screen) and "doubling the tube does not change the separation" (it does — 0.089 to $0.179 \mu\text{s}$ in both runs; separation was never the problem).

Teaching limit: Both width rules are *assumptions this activity declares*, not measurements, and the resolution criterion is a stated classroom rule. Clearing it is not a claim that any instrument could distinguish 200 from 205 Da, nor that it could identify either.

Misconception: a longer flight tube always means better resolution. Only if the peak width does not grow with it. Run the proportional-width assumption and the 2.00 m tube is no better than the 1.00 m tube. Which rule describes a real instrument is a question about that instrument, and this activity assumes rather than measures it.

Model and boundaries

$$qV = \frac{1}{2}mv^2 \rightarrow t = L \sqrt{m / (2qV)}$$

- t is field-free flight time in seconds; L is length in metres; m is mass converted from Da to kilograms.
- $q = z \cdot e$ is positive charge in coulombs; z is the charge number; V is voltage in volts.
- The clock begins at drift-tube entry. **Acceleration time is excluded.**
- The Gaussian peak uses σ as its timing spread, $\text{FWHM} = 2.355\sigma$, and area proportional to packet count.
- Stages 1–3 give each packet a single arrival time: there is no spread in starting position or speed. Stage 4 is the only place a width exists, and there it is assumed.

This ideal linear TOF does not model an ion source, extraction optics, reflectron focusing, detector efficiency, calibration, fragmentation, measurement uncertainty, or molecular identification. Full per-stage assumption lists are in the app's

Details drawer, which is the surface students should be sent to when they ask "but is that real?".

Discussion questions

- Which values on screen are controlled inputs, and which are detector observations? How can you tell from the labels?
- Why does the +2 ion arrive first when mass is held constant?
- How can two different ions make one peak in stage 3?
- What additional evidence would you need before naming a real molecule?
- In stage 4, which assumed width rule lets the longer tube cross the stated criterion, and why does the other one not?
- The page labels some things Live and some Scripted. Pick one of each and say what the label is promising you.

Presenter controls

The four stage tabs run across the top; stages 2 and 3 unlock as the previous stage produces evidence, and stage 4 is open from the start. **Details** in each stage intro opens that stage's formula, assumptions and sources. **?** reopens the Guide. **⚙** Settings offers **Open Presenter Notes** (this document), **Presentation mode** and **Reset**.

Presentation mode scales the page up one notch, widens the instrument, and adds a **Notes** button to the header for these cues. Reset returns the demo to a fresh load and *leaves presentation mode as you set it*. Run locks the prediction during flight; Replay uses the same physical-to-screen scale, so two runs are directly comparable. A stage can be deep-linked with #stage-4.

Keep inputs and evidence distinct. The ion cards show values the simulation already knows and are labelled **Scripted**. Exact arrivals and the completed trace stay hidden until Run and are labelled **Live**. The navy line is the summed detector signal; coloured ticks identify the known packet centres. A peak represents a group arriving over a timing range, not one ion's path.

Sources and attribution

- Iowa State University Chemical Instrumentation Facility, [Mass Spectrometry Tutorial](#) — ions are accelerated into a field-free drift region with kinetic energy qV , and lighter ions reach the detector sooner.
- Chemistry LibreTexts, [Time-of-Flight](#) — the same flight-time relation, and that ions drift through the field-free region at the velocity reached during acceleration, in a tube of one to two metres. This demo's 1.50 m tube is in that range.
- Physical constants from NIST: [elementary charge](#) (exact by SI definition) and [the atomic mass constant](#).

These sources support the underlying physics; they do not endorse this activity. A Murray State project record supports plausible course fit; it is not named here, and the demo is not presented as any instructor's instrument or assignment.

Time-of-Flight Mass Spectrometer · Bryant Harrison · Murray State University · Offline teaching model. Open `index.html` to run the investigation. `src/demo-guide.html` is the canonical source of this document; the shipped `teacher-guide.html`, the PDF beside it, and the in-app presenter notes are all generated from it at build time.